

Predicting Nasdaq Prices with Machine Learning

Data Science Final Project

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Project Overview

Nasdaq Marketplace

It is a stock index for technology leading companies. Nasdaq hosts leading technology companies. It drives innovation and growth.

Predictive Aim

Our project predicts Nasdaq price future delivery. We apply machine learning techniques. This uncovers market dynamics.

Project Overview



Goal

Predict Nasdaq prices with ML



Data Source

MT5 CSV export



Stages

Preprocessing to prediction and evaluate

Our project aims to build an accurate machine learning model for Nasdaq price prediction, leveraging historical data from MT5 and following a structured data science workflow.



Trading Strategy Overview

Model Type

The model is a "buy model" which we only take longs.

Prediction Trigger

We assign the probability of whether we will be pricing higher of 0.1% from now which is predicted by the trained model. If this probability is higher than 60% we take a trade based on that.

Risk Management

We place stoploss 0.2% lower and take profit for 0.3% higher.



Data Collection and Preprocessing and Visualization



Data Source

Historical Nasdaq prices from
MT5



Preprocessing

Handling missing values,
cleaning, normalization

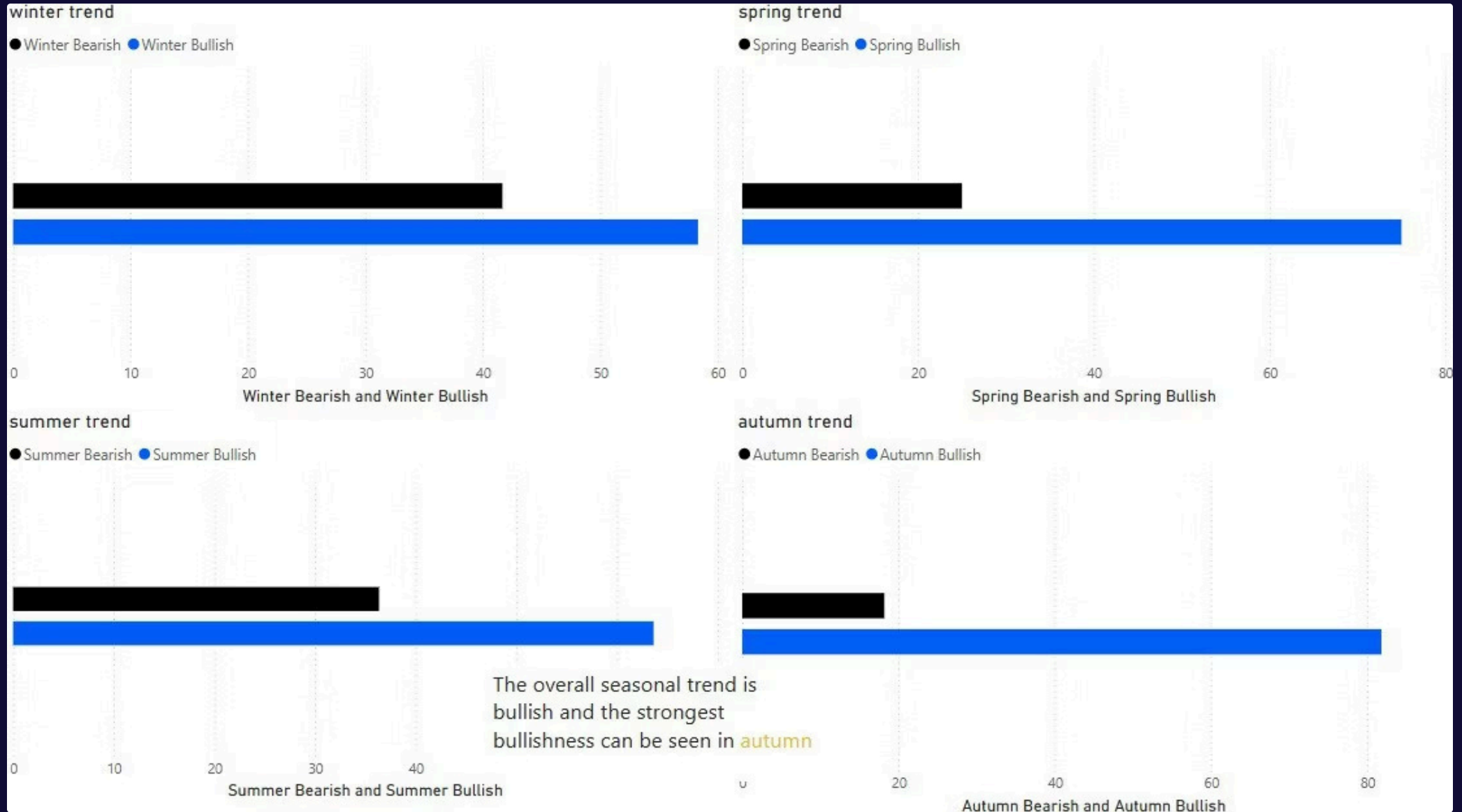


Visualization

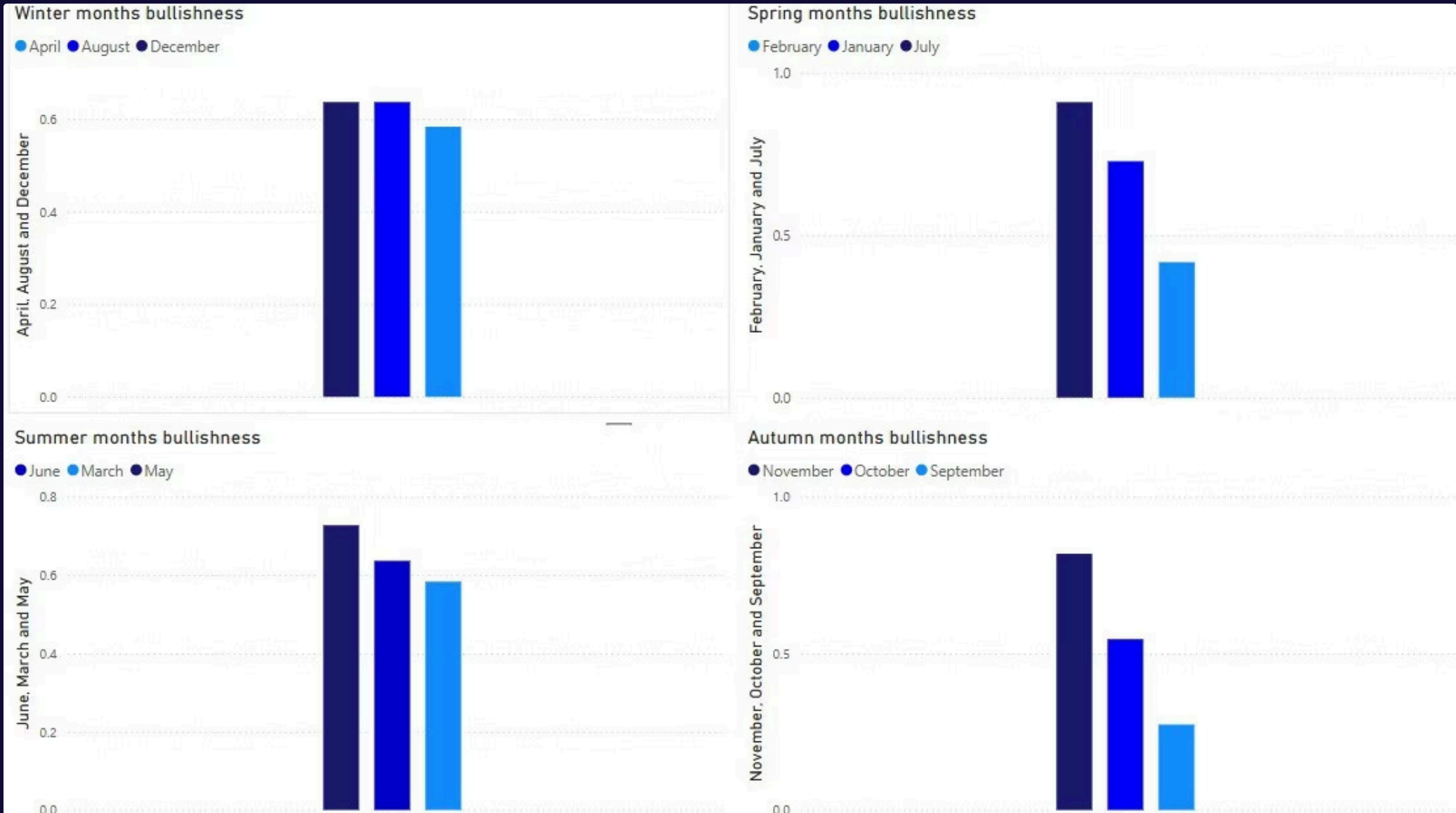
Insights and Statistics

We collected comprehensive historical Nasdaq data. Our preprocessing focused on ensuring data quality, handling any gaps, and normalizing values for consistent model input.

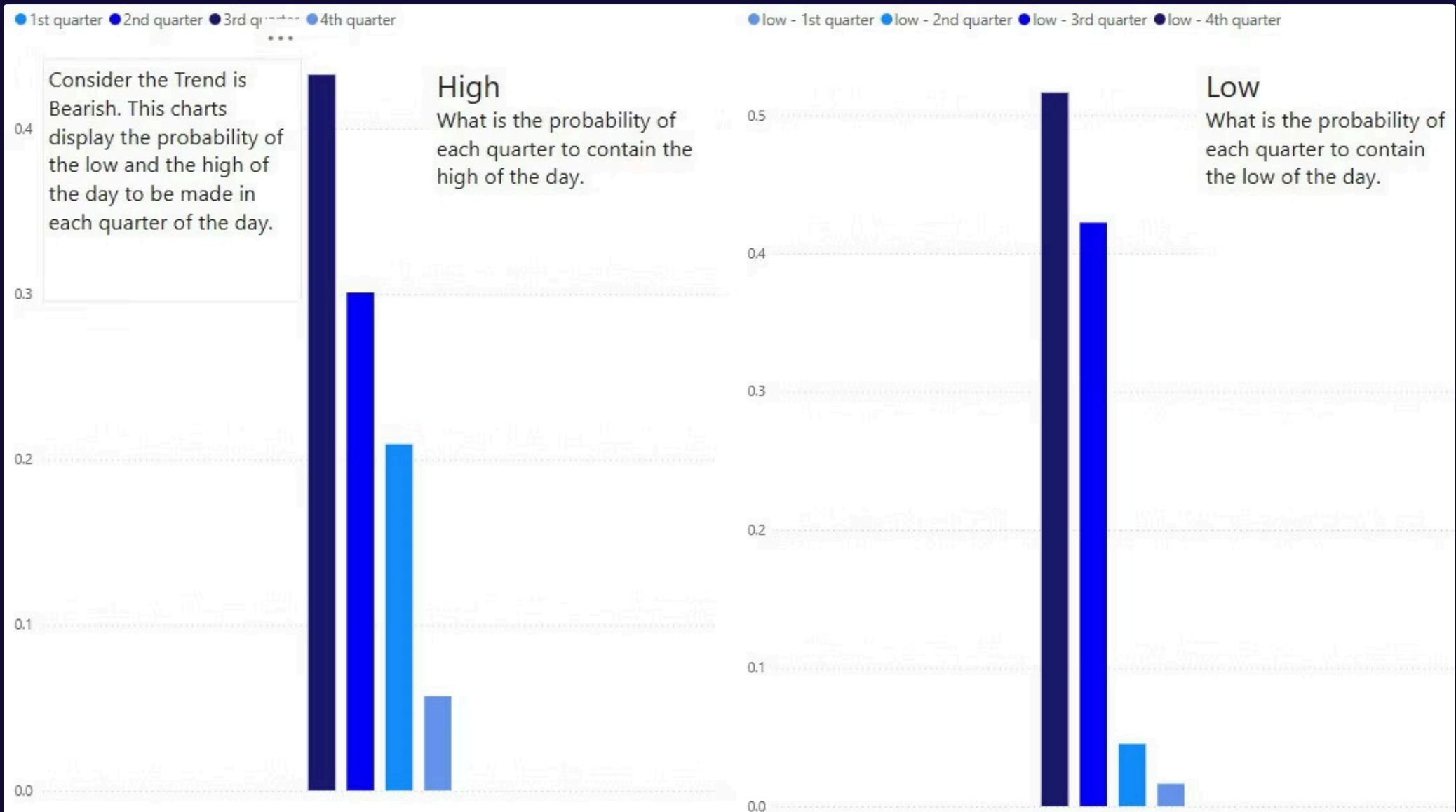
Visualization



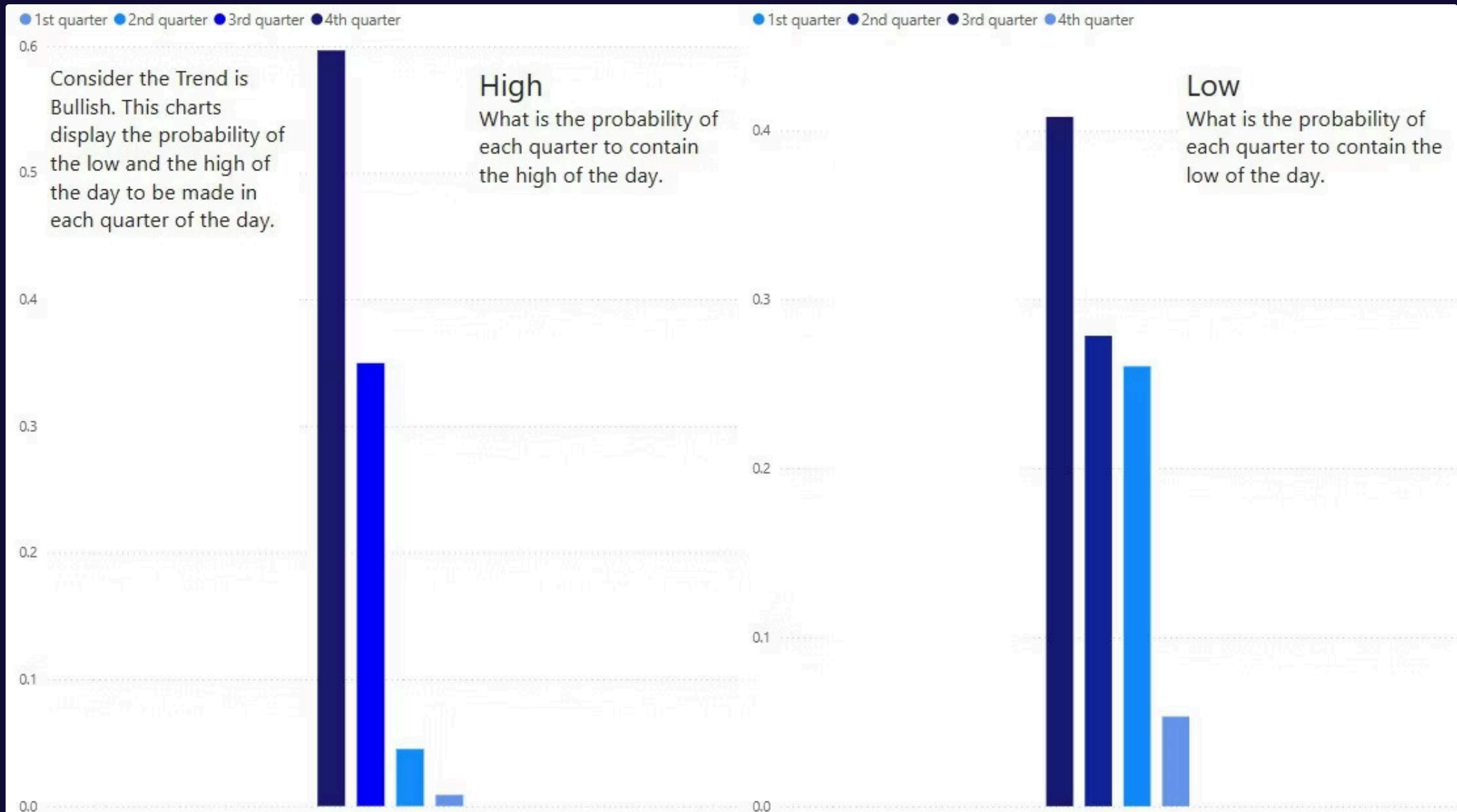
Visualization



Visualization



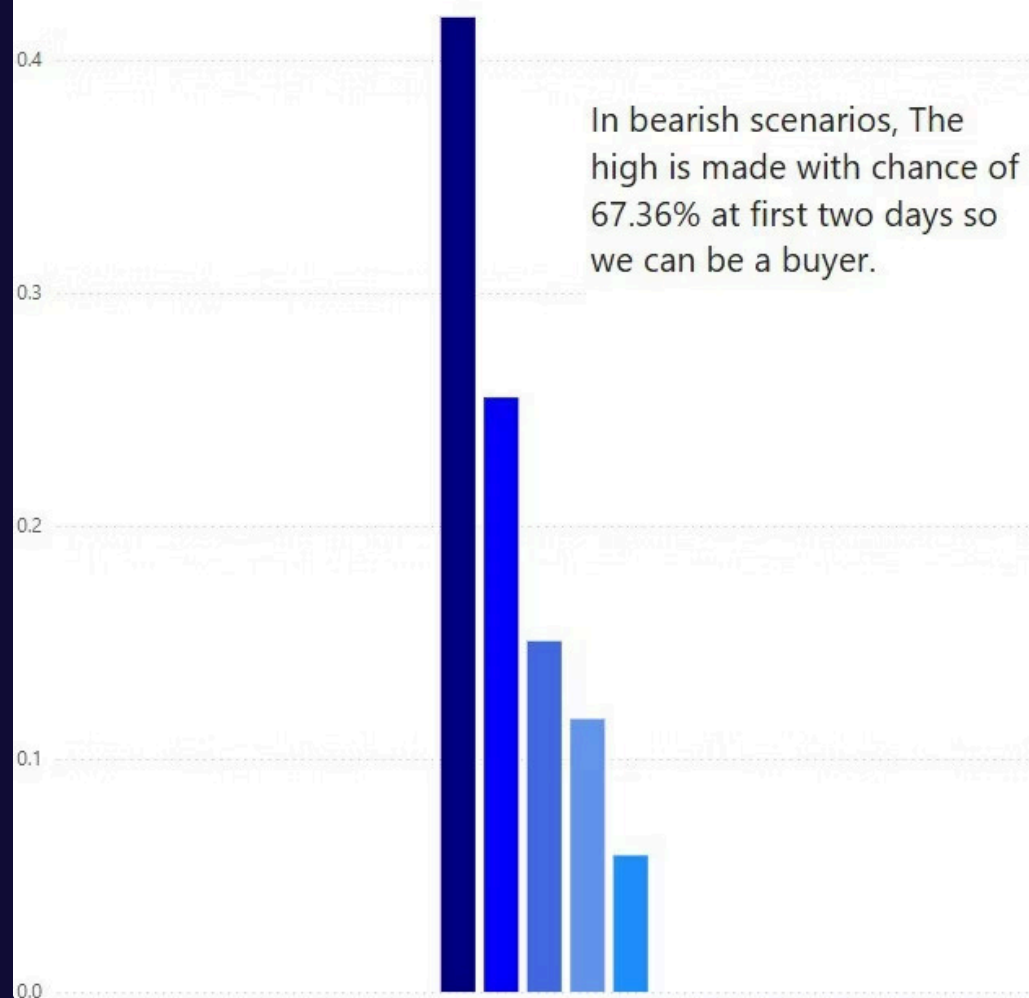
Visualization



Visulization

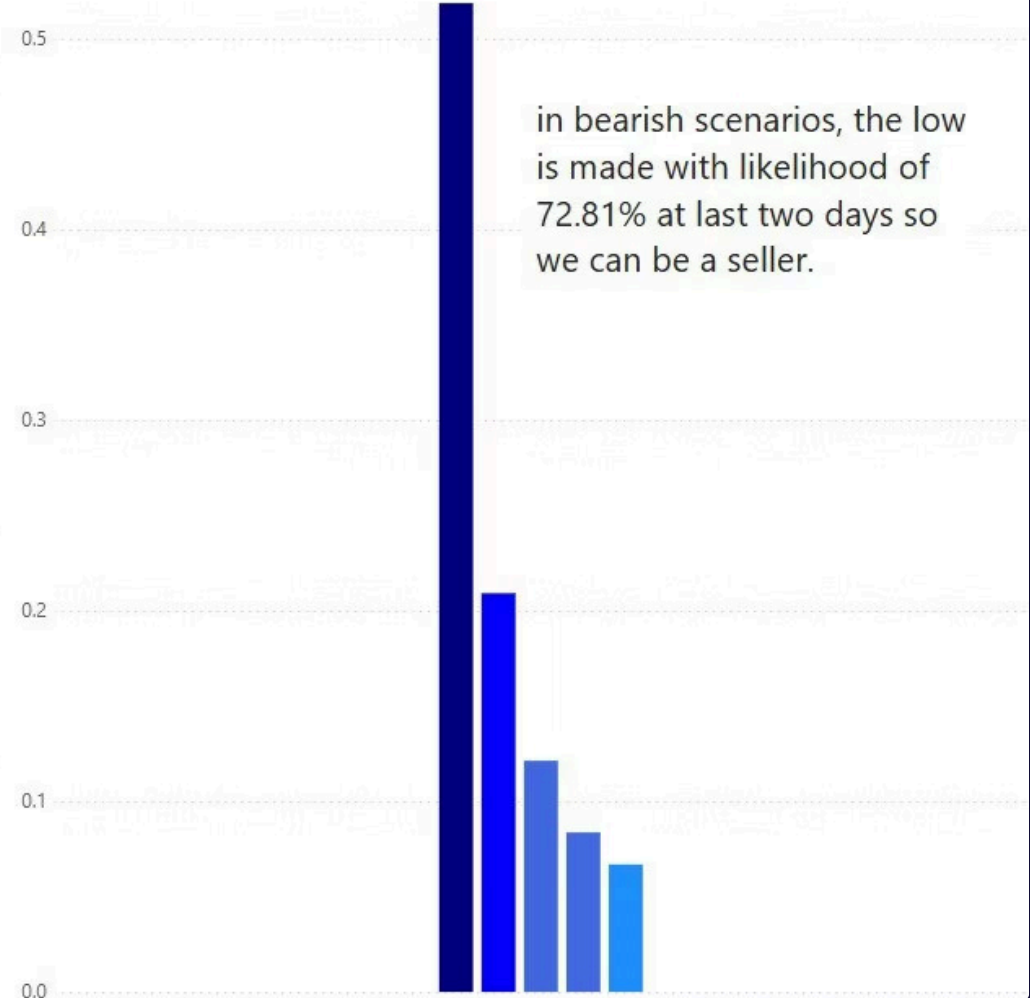
High being made in Bearish scenarios

Monday - high Tuesday - high Wednesday - high Thursday - high Friday - high



Low being made in Bearish scenarios

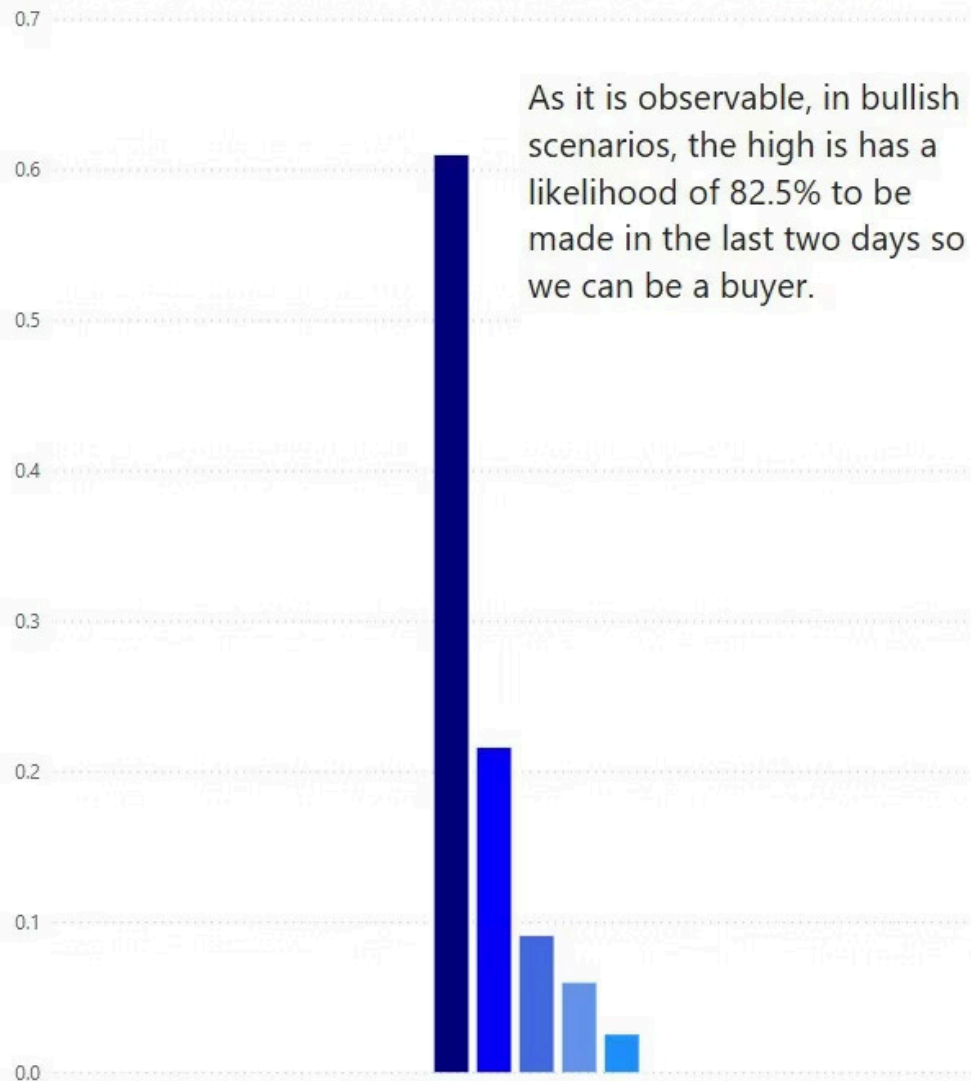
Monday - low Tuesday - low Wednesday - low Thursday - low Friday - low



Visualization

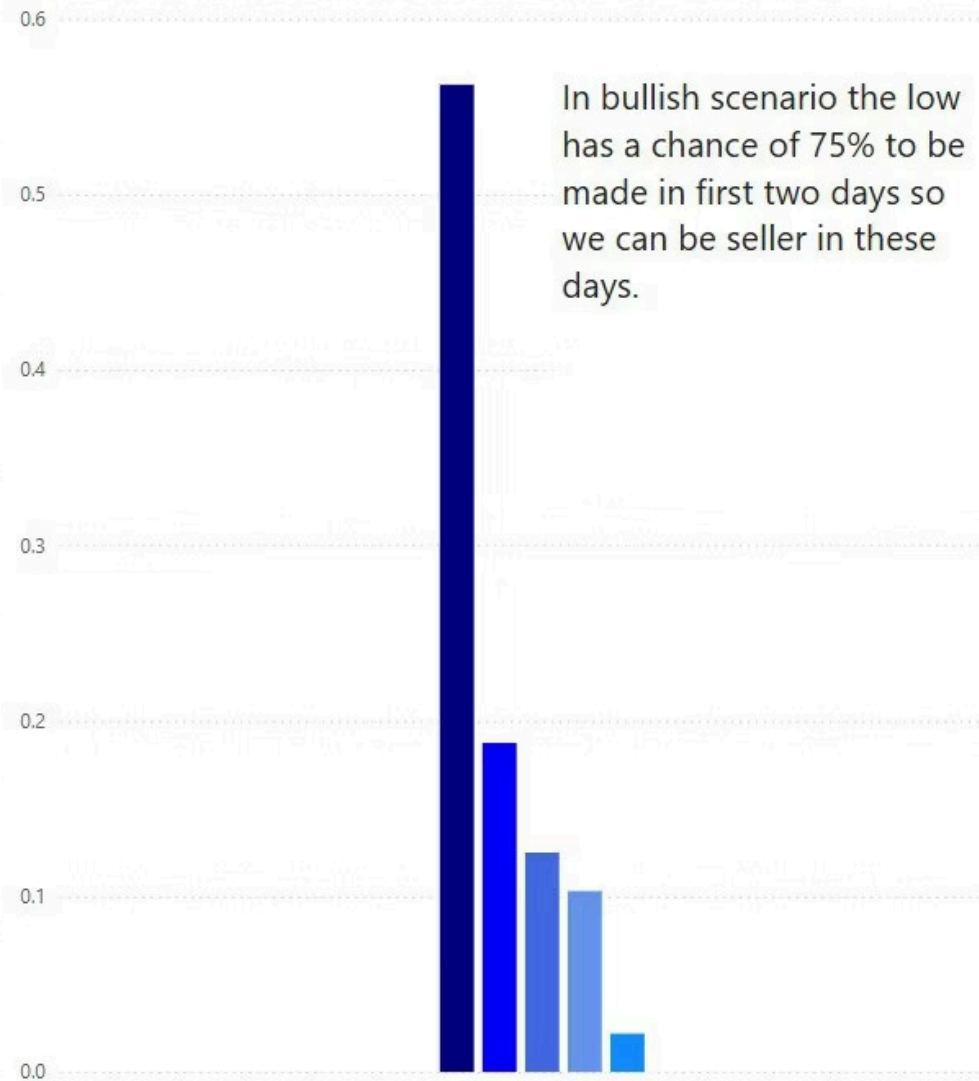
High being made in Bullish scenarios

Monday - high Tuesday - high Wednesday - high Thursday - high Friday - high



Low being made in Bullish scenarios

Monday - low Tuesday - low Wednesday - low Thursday - low Friday - low



Feature Engineering

Technical Indicators

- Simple Moving Averages (SMA)
- Relative Strength Index (RSI)
- Exponential Moving Average (EMA)

We engineered a robust set of features from raw price data, including key technical indicators to capture market dynamics effectively.

Model Selection

Models Considered

- Linear Regression
- Random Forest

Final Model

Random Forest Regressor

Reason for Selection

Superior performance, robustness, and handling of non-linear relationships in financial data.

We evaluated multiple powerful models, ultimately selecting Random Forest for its proven ability to handle complex, noisy datasets and deliver strong predictive results.



Performance Metrics

We evaluated our machine learning model using standard regression metrics to measure its predictive performance.

These parameters are for when probability of profit is above 60%.

Precision: Precision indicates the fraction of true positive predictions among all positive predictions made by the model.

Recall: Recall measures the fraction of true positive instances that were correctly identified by the model.

F1-score: The F1-score is the harmonic mean of precision and recall, providing a balanced measure of model performance.

Accuracy: Accuracy represents the percentage of correct predictions made by the model out of all predictions.

Performance Metrics

First column is for when 60% condition meets. when probability of profit is above that. when it is below 60% metrics are in second column.

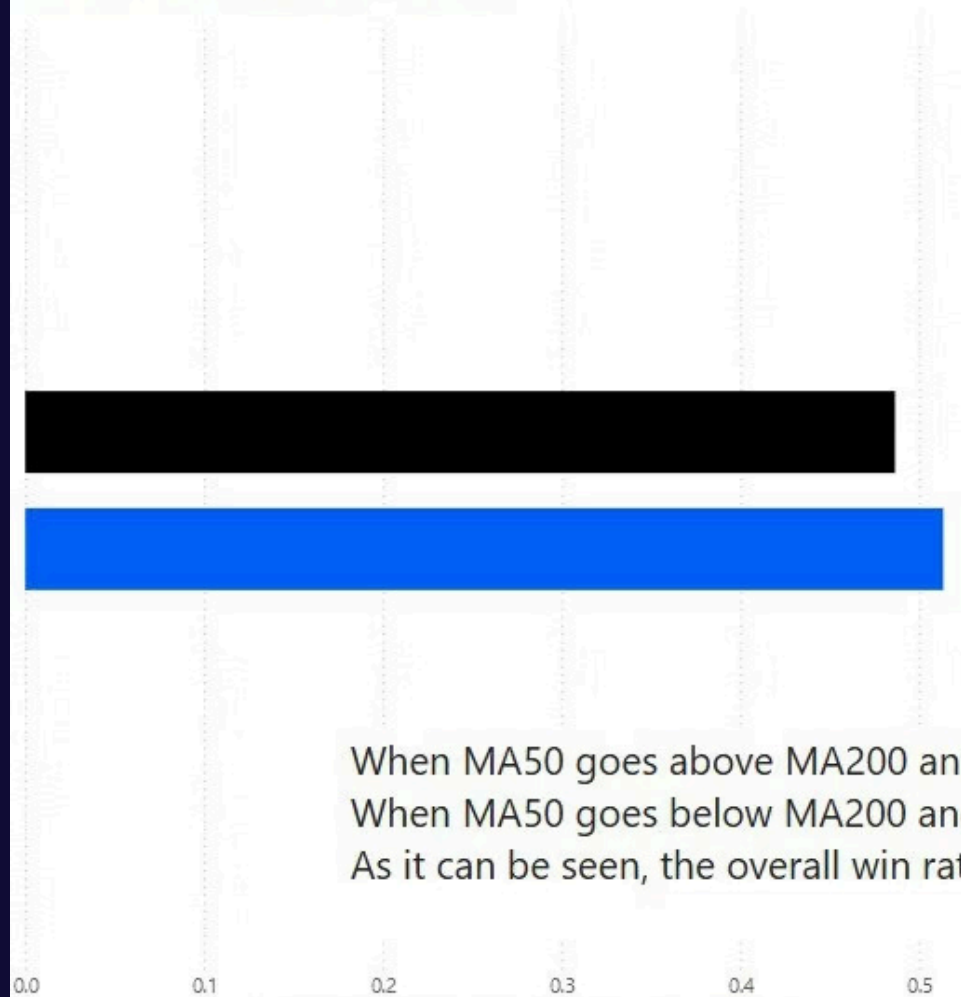
Precision	86%	52%
Recall	96%	25%
F1-score	91%	34%

Accuracy	57.81%
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Performance Metrics

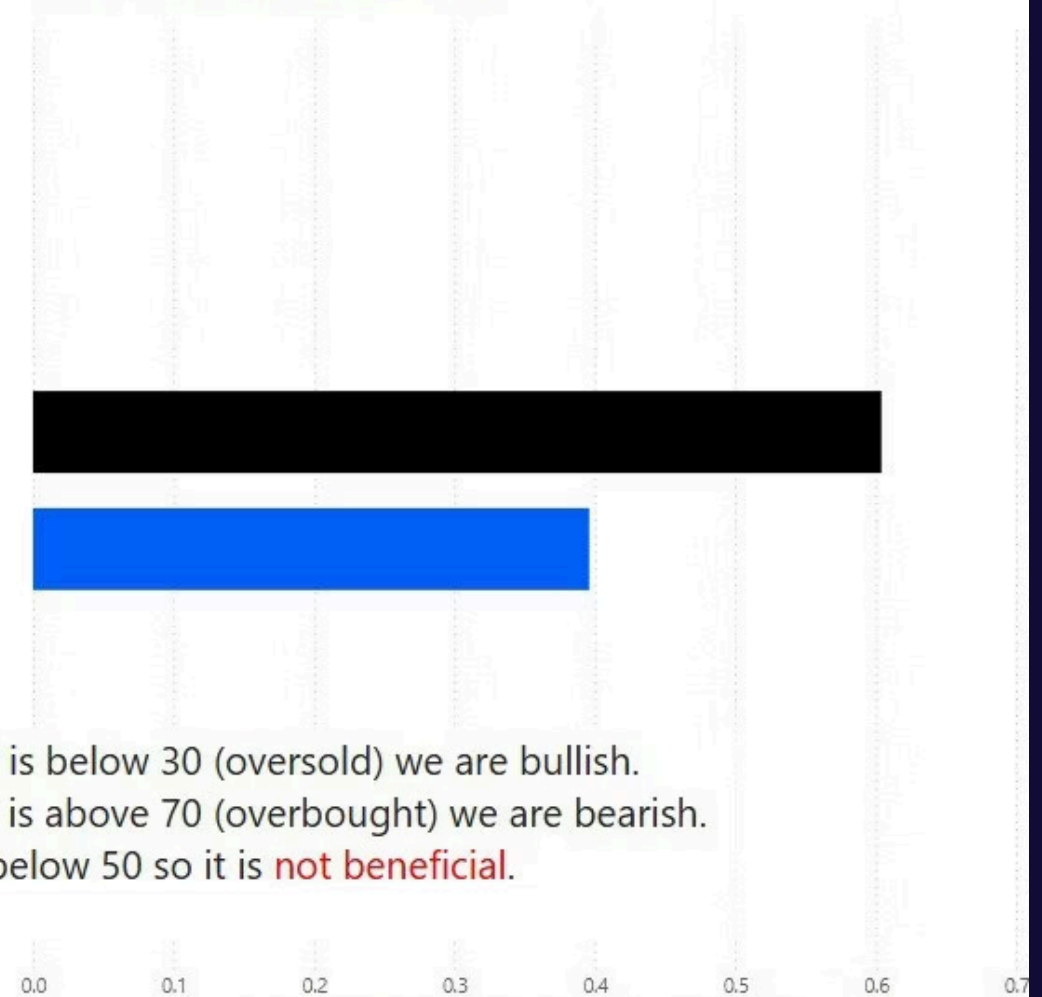
Bullish winrate using RSI and MA50 and MA200

● lose rate - RSI below 30 ● win rate - RSI below 30



bearish winrate using RSI and MA50 and MA200

● lose rate - RSI above 70 ● win rate - RSI above 70



When MA50 goes above MA200 and RSI is below 30 (oversold) we are bullish.
When MA50 goes below MA200 and RSI is above 70 (overbought) we are bearish.
As it can be seen, the overall win rate is below 50 so it is **not beneficial**.

lose rate - RSI below 30 and win rate - RSI below 30

lose rate - RSI above 70 and win rate - RSI above 70

Conclusion:

Due to the complexity of the stock market in its whole, it is not easy to build a very highly accurate trading model. This machine learning based model is not a very high accurate model but yet long term profitable if we use impeccable risk management. Most of the big companies are not trading for large amounts of profits (percentage based) yearly and many of them are not highly accurate in their transactions but yet profitable due to their risk management strategies.

Thanks for your attention !